

Oral Soft-Tissue OCT Research Direction: A Measurement Layer for a Future Clinical Decision Support System

Prepared for concept development and data-partner discussions

Literature window: reviewed through 25 September 2026

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Bottom line

Yes—pursue this direction, but reposition it. There is a credible and potentially publishable contribution at the intersection of oral OCT, scarce-label learning, measurement quality control, and uncertainty quantification. The opportunity is **not** to claim the first automated oral-OCT segmentation system, the first attenuation-based oral biomarker, or the first low-label oral-OCT method. Those components already have meaningful precedents. [1](#) [2](#) [3](#)

The strongest first-paper contribution is a **quality-gated, patient/session-volume-level measurement method** for oral epithelial thickness that supports a later clinical decision support system (CDSS). The method should estimate the epithelial surface and visible epithelial–stromal boundary (ESB), return an explicit *measurement unavailable / boundary not visible* state when appropriate, express thickness in physical units under a stated refractive-index convention, and attach a calibrated uncertainty interval. Its purpose is to make OCT measurements reviewable and trustworthy under constrained data access—not to diagnose cancer, replace biopsy, or make autonomous clinical recommendations.

Recommended first-paper claim: A prespecified OCT measurement procedure can provide reviewable oral epithelial-thickness estimates, quality-gated abstention, and patient-level calibrated uncertainty within a defined acquisition setting, while testing whether selected physical/acquisition features and reduced-label learning improve a matched baseline.

That is an original and useful contribution **if it is empirically earned through patient-level testing, transparent failure analysis, and a held-out calibration set.** It creates the first safe building block for a CDSS: reliable quantitative documentation and longitudinal comparison, rather than an unsupported lesion classifier.

1. What the literature already establishes

The novelty baseline is higher than the original concept note assumed. Oral OCT has been used to measure epithelial thickness across normal oral sites, and thickness is strongly site-

dependent; a single universal normal value is not appropriate. 4 Earlier work also analysed the epithelium–lamina propria boundary in oral precancer imaging. 5

Three recent studies are especially important comparators. Hill *et al.* developed an in-vivo oral OCT pipeline involving field-of-view and artefact detection plus surface and ESB segmentation. Their annotated dataset contained 294 images from 60 patients, and the reported test Dice was 0.98 for the surface and 0.83 for the ESB. 1 Malone *et al.* subsequently used segmentation-assisted morphology, ESB visibility, and attenuation-derived features in 70 sites from 40 patients. 2 Liao *et al.* reported semi-supervised joint segmentation and denoising, including pseudo-label transfer from skin OCT, in healthy oral scans. 3

These studies mean that the following statements should be removed from a proposal or abstract:

- “First oral OCT segmentation method.”
- “First epithelial-thickness measurement method in oral OCT.”
- “First attenuation-aware oral OCT analysis.”
- “First low-label oral OCT learning or skin-to-oral transfer.”
- “No public oral OCT dataset exists.”

The final statement should instead read: **“We did not identify an openly downloadable, reusable in-vivo oral soft-tissue OCT dataset with the necessary clinical or histopathological labels. Important contemporary cohorts are non-public or available only by request; this is an access-constrained-data finding, not proof that no dataset exists anywhere.”** 1 2 3

The potential white space is the **integrated evaluation standard**: patient-level rather than image-level units; an explicit unmeasurable-boundary state; a traceable acquisition-quality gate; physical-unit measurement; label-budget comparisons with leakage controls; and split-conformal thickness uncertainty whose assumptions are stated plainly. The review identified no oral-soft-tissue OCT study that clearly combines all of these elements with patient/session-volume-level calibration. This is an absence-of-findings statement, not an absolute priority claim. 6

2. A publishable research question and hypotheses

Primary research question

In scarce, guarded in-vivo oral OCT data, can a prespecified acquisition-geometry- and quality-conditioned two-boundary model provide patient/session-volume-level

epithelial-thickness measurements with appropriate abstention and calibrated uncertainty while retaining performance at reduced manual-label budgets?

The question is deliberately about a **measurement process**, not a diagnostic label. It is feasible with annotated OCT volumes and acquisition metadata, and it does not require an immediate histopathology-linked cancer cohort.

Primary hypothesis: quality-conditioned measurement

H1. Relative to an otherwise identical structural two-boundary baseline, a model using prespecified acquisition/geometry information and a quality-validity mask will reduce ESB position error and false measurement production in the predeclared deep, low-signal, artefact, or boundary-ambiguous strata, without worsening overall patient-level thickness error.

The primary endpoints should be ESB physical position error, epithelial-thickness mean absolute error and bias, correct *measurable versus unmeasurable* classification, and human-rater disagreement. Dice should be secondary; it can conceal clinically relevant displacement of a thin boundary. A physical “attenuation” term should be called an **effective, system- and protocol-specific signal-decay descriptor**, not an intrinsic tissue optical property. 2 7 8

Secondary hypothesis: label efficiency

H2. At prespecified patient/volume-level label budgets, a proposed reduced-label strategy will improve the primary measurement endpoints relative to a structural model trained with the same target labels and relative to a strong target-only self-/semi-supervised baseline.

This should be tested at nested budgets expressed in **patients and representative volumes**, not B-scans. A reasonable planning series is 5, 10, 20, 40, and 60 annotated volumes, repeated within the development cohort while retaining a locked patient-level test set. Any observed improvement must be estimated with uncertainty across repeated budget samples. Existing semi-supervised oral work makes this a comparative question, not an inaugural claim. 3

Secondary hypothesis: useful conformal uncertainty

H3. With a frozen model, score, thickness definition, and quality gate, a split-conformal procedure calibrated on held-out patient/session-volumes will attain the prespecified **marginal** coverage target for consensus-reference thickness within the stated eligible acquisition stratum, and its interval widths will be reported against a prospectively defined utility criterion.

The wording matters. Conformal prediction provides finite-sample distribution-free *marginal* coverage under exchangeability of calibration and test units; it does not guarantee correctness for an individual patient, conditional subgroup coverage, or robustness after uncalibrated device/site shift. 6 9 Calibration therefore belongs at the patient/session-volume level, not the B-scan level. A new device, oral site, operator population, or acquisition protocol requires a separate calibration reserve or a separate, justified shift analysis.

Optional hypothesis: cross-tissue transfer

H4 (only if data-use rights permit). At the same target-data label budget and with the same backbone/fine-tuning protocol, a specified donor representation source will outperform random initialisation and be compared with target-only self-supervision, ImageNet, and a size-matched large but anatomically mismatched OCT source.

This is the corrected version of the original “physical similarity beats data size” hypothesis. It should not presuppose the result. Human oesophageal epithelial OCT is a more coherent documented mucosal source than retina, while skin is useful as an acquisition-statistics hypothesis but is not anatomically interchangeable with oral mucosa.

10 3 If only a much larger source wins, the conclusion is **a scale advantage**, not evidence against tissue matching.

3. Method design that can support these claims

The model should produce two ordered traces: the tissue surface and the visible ESB. It should also predict a separate status: **measured**, **ambiguous**, or **unmeasurable**. An absent or indistinct ESB must never be interpreted automatically as invasion, inflammation, thickness, or a negative result; it may result from tissue state, contact, focus, low signal, shadow, saliva, motion, or field-of-view failure.

Thickness should initially be axial thickness because it is easier to specify and reproduce. The protocol should state the optical-to-physical depth convention, the refractive index used, the region-of-interest rule, and a sensitivity analysis for that refractive-index assumption. It should preserve two independent expert annotations on a challenging subset; consensus is the reference target, while disagreement measures the irreducible uncertainty of the reference itself.

The physics contribution should be modest but rigorous. Before training, document the signal format, axial/lateral spacing, wavelength, bandwidth, power, focus/Rayleigh length, sensitivity roll-off, noise floor, saturation behaviour, reconstruction/denoising, and contact protocol. Perform session-level homogeneous and layered phantom checks. Use an attenuation/signal-decay descriptor only in segmentation-supported layer interiors that pass a predeclared validity mask. 7 8

A credible comparator set contains: (1) a conventional structural two-boundary baseline; (2) the same architecture with explicit measurement geometry, boundary ordering, and unmeasurable output; (3) the quality/physics-conditioned variant; (4) target-only self-/semi-supervised learning; and, only if lawful source access exists, (5) donor-pretrained variants. Every model must use identical patient partitions, resampling, augmentation, fine-tuning schedule, and model-selection rule.

All scans, B-scans, patches, repeat visits, contralateral sites, and augmented descendants from one patient must remain in the same partition. The minimum defensible partitions are separate development, calibration, and untouched test cohorts. For a main internal study, planning for roughly 80–120 patients with independent calibration and test reserves is sensible; a 40-patient pilot can establish annotation and quality-control feasibility but should not claim robust subgroup or deployment performance. These are planning estimates, not formal sample-size calculations.

4. How the CDSS changes the aim—without inflating the first paper

The **end goal** should be an OCT-based CDSS. The first study becomes more valuable when it is explicitly positioned as the CDSS’ s **measurement, quality-control, and uncertainty layer**.

The initial CDSS should be a clinician-reviewed *measurement-and-documentation aid*. It should retain the raw OCT volume, acquisition metadata, proposed boundaries, editable overlays, quality reasons, physical measurements, uncertainty, and audit trail. It may help document a scanned region and compare technically comparable scans over time. It must not call a scan benign, malignant, normal, negative, safe to monitor, or biopsy-free.

A suitable research intended-use statement is:

For trained clinicians using specified OCT hardware and acquisition protocols, the software displays source OCT images; proposes the epithelial surface and visible epithelial–lamina propria boundary; and reports reviewable thickness and prespecified attenuation-related measurements with quality and uncertainty status. It supports documentation and technically comparable serial review. The clinician reviews the raw image, overlays, patient context, and local pathway before making any referral, biopsy, diagnosis, or treatment decision. The software does not diagnose or exclude disease, grade dysplasia, determine whether a biopsy is required, replace histopathology, or independently make a clinical decision.

This distinction is scientifically and translationally important. Imaging software used in a clinical context may be regulated based on its **specific function and intended use**, not on whether it is called a “CDSS.” A future deployable product therefore needs jurisdiction-

specific regulatory assessment, lifecycle controls, human-factors work, and prospective evidence. ¹¹ ¹² ¹³

Programme phase	Near-term deliverable	Allowed conclusion
1. Analytical measurement validation	Boundary/thickness agreement, repeatability, quality-gate performance, uncertainty calibration, patient-level test	The system measures its defined OCT quantities within its tested scope.
2. Clinical association study	Prospectively mapped OCT–pathology or clinical association design	A defined reviewed feature is associated with a defined reference standard.
3. Silent-mode and usability study	Completion, rejection/edit patterns, error modes, clinician understanding, workflow burden	Clinicians can use and understand the reviewed outputs in the intended workflow.
4. Comparative impact evaluation	Predefined documentation, pathway, or communication outcomes plus harms	The exact CDSS workflow produces the evaluated process effect in the studied setting.
5. Lifecycle monitoring	Version control, drift indicators, quality/override/error monitoring	The locked version remains within its documented evidence envelope.

This staged structure follows the distinction between analytical validation, clinical association, and clinical validation in SaMD frameworks. ¹² Early prospective CDSS evaluation should assess human–AI interaction and safety-related workflow behaviour rather than infer benefit from an offline segmentation score. ¹⁴

5. What to request from guarded-data partners

Approach collaborators with a **protocol and governance offer**, not a request to export a dataset. The narrow first task needs:

1. **OCT data:** raw linear or minimally processed volumes where possible, plus derived B-scans and the reconstruction configuration. Vendor-rendered images alone may support boundary learning but make physical attenuation claims much weaker.
2. **Acquisition metadata:** device/probe/lens, software version, wavelength, spacing, focus/roll-off information, power, contact/pressure protocol, operator, date/session,

oral sub-site, orientation, and quality issues.

3. **Reference labels:** independent surface and visible-ESB traces; a quality-defect taxonomy; and separate labels for *ESB not visible*, motion, shadow, saturation, poor contact, and insufficient field of view.
4. **Clinical and pathology context:** structured lesion/site mapping and pathology only when a specimen is explicitly mapped to the scanned region. Pathology is a later clinical-association reference, not pixel-level “ground truth” unless co-registration, timing, and deformation are handled.
5. **A partner-held test route:** a patient-level frozen test cohort that the model developers cannot inspect during tuning, ideally evaluated locally through a locked container or secure enclave.

A collaboration charter should define the approved purpose, consent/ethics route, roles, access, data/model ownership, publication rights, commercial-use boundaries, site withdrawal, model update approval, incident handling, and audit rights. A controlled enclave or site-run evaluation container often provides more scientific credibility than “data available on request.” Federated learning can be a later option, but it does not remove privacy, governance, leakage, heterogeneity, or local-validation requirements. 15

6. Novelty statement for a proposal or journal introduction

Use this wording, revising only after the final literature update at submission:

Automated oral OCT boundary segmentation, attenuation-derived biomarkers, and sparse-label learning have each been reported. This study evaluates a distinct measurement-focused integration for access-constrained in-vivo oral OCT: a quality-gated two-boundary estimator with an explicit unmeasurable-boundary state, acquisition-conditioned physical measurement, patient-level label-budget evaluation, and split-conformal epithelial-thickness uncertainty calibrated on held-out patient/session-volumes. We test the contribution against matched structural and target-only low-label baselines and report the conditions under which the system abstains. 1

2 3 6

That novelty claim is **moderate to strong** if the study includes the predeclared calibration and leakage controls. It becomes weak if it is only another U-Net trained/tested on correlated B-scans, or if uncertainty is merely a visual heat map without a calibration/test protocol.

7. Appropriate first-paper positioning

A strong primary target is **Biomedical Optics Express** when the study includes genuine acquisition characterisation, phantom checks, and a controlled physics/quality ablation. A more workflow- and reproducibility-focused manuscript could fit **Computer Methods and Programs in Biomedicine**. More selective general medical-image-analysis venues should be treated as later targets unless the method shows true independent device/centre validation.

Possible titles are:

- **Quality-gated estimation of oral epithelial thickness from OCT with patient-level uncertainty reporting**
- **Reliable oral mucosal epithelial measurement from guarded OCT data: quality control, label efficiency, and conformal uncertainty**
- **An uncertainty-aware measurement layer for oral OCT clinical decision support**

Avoid putting “cancer detection,” “biopsy guidance,” “diagnosis,” “cross-device,” “device-independent,” or “guaranteed clinical confidence” in the first paper’s title or abstract unless a separate study has earned those claims.

8. Immediate actions

1. **Confirm the data reality.** Identify device(s), raw-data export, acquisition protocol, number of unique patients and sessions, sites, normal/lesion distribution, and whether an independent expert annotation process is possible.
2. **Write the annotation and quality manual before modelling.** Define surface, ESB, unmeasurable ESB, artefacts, physical conversion, region of interest, and adjudication.
3. **Secure the split before labels are expanded.** Freeze patient-level development, calibration, and test membership; reserve a partner-held external test option if possible.
4. **Run a 10–20 patient technical pilot.** Assess raw-data suitability, ESB visibility, rater agreement, quality-failure rate, phantom stability, and whether a measurement is sufficiently repeatable to justify the main study.
5. **Choose the core paper deliberately.** WP1 quality-controlled measurement plus WP2 patient-level conformal uncertainty is the highest-value core. Add label efficiency next. Add attenuation or cross-tissue transfer only when the necessary calibration/source access is secured.
6. **Design the CDSS in parallel but deploy nothing clinically.** Create a low-fidelity clinician-review workflow showing raw image, traces, uncertainty, quality reasons, edits, and serial-comparison limitations; use it to shape the later silent-mode study.

Conclusion

The proposed direction is worth pursuing. Its real value is not a broad promise that physics-informed AI will solve oral cancer diagnosis with scarce OCT data. It is a tighter and more defensible programme: **make oral OCT measurements auditable, quality-aware, and uncertainty-calibrated first; then build clinical association and decision-support evidence in stages.**

This approach is both publishable and strategically aligned with an eventual CDSS. It gives data custodians a responsible collaboration proposition, gives reviewers a falsifiable methods contribution, and gives clinicians a tool that is designed to disclose its limits rather than conceal them.

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